A Summary and Discussion of *Exploring Human-Like Translation Strategy with Large Language Models*

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Abstract

This report reviews *Exploring Human-Like Translation Strategy with Large Language Models* by He et al. (2024), published in *Transactions of the Association for Computational Linguistics*. The paper introduces Multi-Aspect Prompting and Selection (MAPS), an inference-time framework where a large language model generates keyword, topic, and demonstration guidance, uses each type to produce a separate translation candidate alongside a direct translation, and selects the best candidate through quality estimation. Across three models and 11 WMT22 directions, MAPS outperforms ordinary reranking under an equally sized candidate pool, with an average COMET gain of about 0.7 points and hallucination reductions of up to 8%, compared to 3% for reranking. I summarize the method and experiments, and then discuss how far MAPS's design is genuinely human-like rather than merely human-inspired engineering, the risk that its three guidance types share the same underlying model errors, and its dependence on imperfect candidate selection.

1 Introduction

Large language models can produce fluent and grammatically well-formed translations, but fluency does not guarantee that a translation is correct. A natural-sounding output may select the wrong sense of an ambiguous expression, get specialised terminology wrong, omit source information, or introduce unsupported content. Translation performance also varies across language directions and resource conditions (Hendy et al., 2023). These limitations suggest that a direct instruction to translate a sentence may not fully use the linguistic and analytical information represented in an LLM.

Translation is not only target-language generation but also interpretation and decision-making. Human translators may identify difficult expressions, infer a text's subject and purpose, consult terminology or comparable examples, consider alternatives, and evaluate a formulation before accepting it. These activities are not fixed or universal, varying with the translator, language pair, and purpose. Nevertheless, preparation, consultation, revision, and evaluation are recognisable elements of translation practice (Bowker, 2002; Pym, 2014).

He et al. (2024) illustrate the interpretive problem with the expression "Hallucination issue in LLM." The abbreviation *LLM* could denote a *large language model* or a *Master of Laws*. A direct translation may be fluent while selecting the wrong meaning. Recognising that *hallucination* is an artificial-intelligence term, or inferring that the sentence concerns language models, makes the intended interpretation more likely. The example motivates a framework in which the model performs explicit preparation before committing to one translation.

Multi-Aspect Prompting and Selection (MAPS) is designed for this purpose. It is an inference-time framework that leaves the model's internal weights unchanged. Before translation, the model generates three forms of guidance: important source expressions with proposed target equivalents, a sentence-level topic, and a related bilingual demonstration. Each type is then used separately to generate a candidate, while a direct zero-shot translation remains available. A quality-estimation procedure selects the final output. MAPS therefore tests whether an LLM can generate useful guidance, whether that guidance improves candidate generation, and whether the system can identify the strongest candidate when guidance is inaccurate.

The term *knowledge* used in the paper requires qualification. MAPS does not verify its keyword pairs against a terminology database, retrieve demonstrations from a trusted corpus, or obtain the topic from document metadata. Its intermediate outputs are model-generated guidance and remain subject to uncertainty and error. Separating the

guidance types and retaining the direct candidate reduces the risk of forcing every generated analysis into the final output.

Generating several candidates also creates an alternative explanation for improvement: MAPS might help only because it provides more outputs from which to choose. The paper therefore compares it with ordinary reranking under equally sized candidate pools. Reranking combines the deterministic direct translation with three additional samples from the same prompt. MAPS combines that direct translation with keyword-, topic-, and demonstration-guided candidates. Both use the same selector, and this comparison controls candidate quantity while testing whether differentiated prompts create a stronger pool than repeated sampling from one generation condition. It does not match total prompting, generated tokens, latency, or computation.

This report examines both the reported quality gains and the mechanism proposed to explain them. Its central position is that MAPS meaningfully reproduces selected procedural features associated with translation practice: preparation, attention to different aspects of the source, consideration of alternatives, and evaluation. However, improved outputs do not show that the model follows human cognitive processes, and sentence-level experiments cannot represent the broader contextual, cultural, interpersonal, and professional dimensions of human translation.

2 Method

2.1 Framework Overview

MAPS reorganises translation into three inference-time stages: Knowledge Mining, Knowledge Integration, and Knowledge Selection. It does not train or fine-tune a new translation model. Instead, it prompts the same LLM several times to generate guidance, produce candidates under different conditions, and select a final translation.

The independence of the three guidance types is important. Topic generation does not depend on the generated keywords, and demonstration generation does not depend on either of the other outputs. Together with the direct zero-shot output, they form the candidate set

$$Y = \{y_{\text{direct}}, y_{\text{key}}, y_{\text{topic}}, y_{\text{demo}}\}.$$

This design creates structured candidate diversity: rather than sampling repeatedly from one prompt, MAPS changes the information supplied before each guided translation. The direct candidate remains an equal member of the pool because generated guidance can make a translation worse.

2.2 Guidance and Candidate Generation

Keyword mining identifies important words or phrases in the source and proposes target-language equivalents. This guidance can focus translation on technical terms or names whose mistranslation would alter meaning. However, the proposed equivalents are not checked against an external terminology resource. Keyword guidance may preserve a crucial term, but it may also impose an unsuitable lexical choice that the model follows consistently.

Topic mining asks for a specific description of the sentence's subject. A focused topic can provide a semantic frame for resolving ambiguity, choosing vocabulary, or selecting an appropriate register. It is more informative than a broad domain label: the *news* domain, for instance, spans subcategories such as political news and entertainment news that call for different registers and tones, a distinction the domain label alone cannot signal (He et al., 2024). However, the topic is inferred from the isolated source sentence, not from preceding text, later paragraphs, author information, or a translation brief. It therefore cannot replace document context and may be speculative when the sentence is underspecified.

Demonstration mining generates a source-language sentence related to the input and its target-language translation, supplied as an in-context example when translating the original source. Unlike conventional few-shot prompting, the demonstration is created for the individual input rather than retrieved from an existing corpus, removing the need for an external datastore but leaving the example unverified: a useful demonstration may clarify a term or provide an analogous construction, while a poor one may reinforce the same error the framework is meant to avoid.

Each guidance type is inserted into a separate translation prompt. Keywords foreground lexical choice, topics foreground interpretation, and demonstrations foreground analogy and in-context learning. MAPS does not assume that one type is universally strongest or that all guidance should be combined. The paper's additional Three-in-One experiment indicates that placing all three types in one prompt can improve a single translation, but adding that output to the MAPS pool produces only

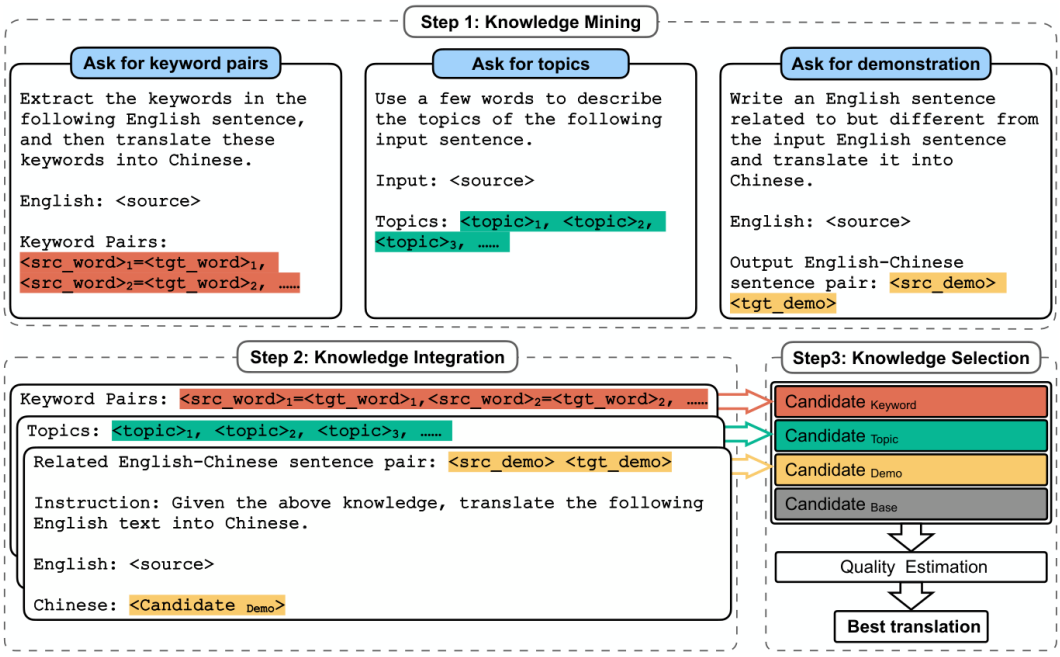


Figure 1: Framework of MAPS. The model generates keyword, topic, and demonstration guidance independently. Each type guides a separate translation candidate, and quality estimation selects the final output. Reproduced from He et al. (2024), Figure 2.

marginal gains, suggesting that it contributes little additional candidate diversity.

2.3 Candidate Selection and Failure Points

MAPS chooses the final translation according to

$$y^* = \arg\max_{y \in Y} Q(x, y),$$

where $Q(x, y)$ estimates the quality of candidate y for source sentence x. Although the stage is called Knowledge Selection, the operational choice is between complete translations. Guidance is accepted or rejected only indirectly through the candidate that incorporates it.

The paper evaluates three selectors. LLM-based Single-Choice Question selection (LLM-SCQ) presents the candidates as options and asks an LLM to choose the best one. This requires no specialised evaluation model but assumes that the LLM can reliably judge its own outputs. COMET-QE uses a reference-free quality-estimation model to score each source–candidate pair and represents the realistic specialised selector in the experiments (Rei et al., 2022b). COMET oracle scores candidates using a human reference and selects the one with the highest reference-based COMET score. It is not deployable or a perfect human selector, but a metric-based upper bound that helps distinguish limitations of the candidate pool from limitations of practical selection.

Failures can arise at two levels: the guidance may be incorrect or push several candidates toward the same error, leaving no strong option in the pool, or MAPS may generate a good candidate but lose it because the selector prefers a fluent yet inadequate output. The framework manages uncertainty through differentiated generation and selection, but does not independently verify guidance or guarantee the best candidate will be recognised.

3 Experiments and Results

3.1 Experimental Setup and Controlled Comparisons

He et al. (2024) evaluate MAPS with text-davinci-003, Alpaca-7B, and Vicuna-7B on 11 directions from the WMT22 General Machine Translation Task, which includes material from news, social media, conversational text, and e-commerce (Kocmi et al., 2022). Translation quality is measured with COMET-22 and BLEURT-20, supplemented by professional preference judgements, Multidimensional Quality Metrics (MQM) annotation, lexical-ambiguity evaluation, and hallucination analysis (Freitag et al., 2021; Rei et al., 2020, 2022a; Sellam et al., 2020).

The deterministic zero-shot translation serves as the baseline. Five-shot prompting supplies labelled examples collected for the relevant direction. Ordinary reranking and MAPS both produce four candidates each, as described above, differing only in how the three additional candidates are generated. The MAPS–reranking comparison is therefore the strongest control for the paper's central claim: it holds the number of candidates and the selection method constant while changing the source of candidate diversity (again, without matching total computation).

3.2 Overall Automatic and Human Results

Across the reported settings, MAPS generally outperforms direct translation and ordinary reranking. With text-davinci-003 and COMET-QE selection, it achieves a higher COMET score than reranking in all 11 directions. For example, English-to-Chinese increases from 86.9 with reranking to 87.6 with MAPS. BLEURT broadly supports the same pattern. These results are consistent, though gains of this size should not be treated as large practical improvements.

Some of the largest gains occur for the weaker Alpaca baseline. On English-to-Chinese, Alpaca rises from 58.9 COMET for direct translation to 66.2 with reranking and 69.0 with MAPS. This pattern is consistent with the interpretation that structured guidance is especially useful when direct translation is weak. However, the study does not manipulate model strength independently, so it does not establish a causal relationship between weaker baseline ability and larger MAPS gains.

Professional human evaluation supports the automatic results for English↔Chinese. In pairwise preference judgements, MAPS wins more often than it loses in most comparisons, although ties remain common. The MQM analysis gives more detailed evidence: for English-to-Chinese, average penalties are 1.59 for MAPS, 1.79 for reranking, and 1.94 for the baseline. For Chinese-to-English, they are 2.60, 2.84, and 2.96. Lower penalties indicate fewer or less severe errors, mainly in mistranslation, style, and omission. This human evidence strengthens the automatic evaluation but directly covers only two of the 11 directions.

3.3 Candidate Pool and Selection

The choice of selector substantially affects the final result. MAPS with LLM-SCQ is approximately comparable to five-shot prompting for text-davinci-003, whereas COMET-QE consistently performs better. Under the tested conditions, a specialised reference-free quality estimator is therefore more reliable than asking the LLM to choose among its own candidates.

Oracle results separate generation quality from selection quality. On English-to-Chinese, MAPS reaches 87.6 COMET with COMET-QE and 88.5 with COMET oracle. MAPS also remains stronger than reranking when both use the oracle selector. These findings indicate both that the MAPS pool contains useful candidates missed by the practical selector and that its pool is stronger than an equally sized reranking pool under the same metric-based selection condition. This remains limited: COMET oracle reflects COMET's own reference-based preference, not necessarily what a professional translator would choose.

The paper also reports a concrete case: for one Chinese-to-English sentence, COMET-QE scored an error-free translation lowest among three candidates, while ranking a translation with fabricated content and one in the wrong target language higher. This is the reverse of the ranking given by the reference-based COMET and BLEURT scores. Selection is therefore not a minor final step. A stronger pool improves the opportunity for a good result, but the final output can still deteriorate if the evaluator rewards surface quality while overlooking adequacy errors.

Candidate frequencies and ablations show why all four generation conditions remain relevant. The direct candidate is selected in approximately 35% of cases, demonstrating that explicit preparation does not improve every sentence. The three guided candidates divide the remaining selections, and replacing any one of them with an ordinary sample reduces average performance. Demonstration removal produces the largest average decline, but the evidence supports complementarity rather than a universally best guidance type.

3.4 Guidance Noise, Ambiguity, and Hallucination

The keyword analysis illustrates both the influence and the risk of model-generated guidance. Source keywords almost always correspond to expressions found in the input, but only 55.8% of generated English-to-Chinese target keywords and 41.8% of Chinese-to-English target keywords occur in the respective reference translations. At the same time, the generated candidates follow those proposed tar-

get terms in 97.1% and 89.5% of cases. Absence from one reference does not prove that a target term is wrong, because valid translations can use synonyms or alternative constructions. Nevertheless, the contrast shows that target-side guidance is less stable than source identification and that the model follows its own suggestions strongly. Selection is therefore a safeguard against influential but unsuitable guidance.

MAPS is also evaluated on a Chinese-to-English lexical-ambiguity dataset introduced by He et al. (2020). Compared with reranking, human-evaluated disambiguation accuracy increases from 61.5% to 65.5%. The gain is moderate, but it directly tests the mechanism proposed by the framework: terminology and topic preparation should help when a correct translation depends on choosing the intended sense of an expression.

Hallucination analysis provides further targeted evidence. For Alpaca's Chinese-to-English translations, ordinary reranking reduces automatically detected hallucinations by 3% relative to the baseline, while MAPS with COMET-QE reduces them by 8%. Human evaluation also reports reductions for Alpaca and Vicuna in both English↔Chinese directions, while text-davinci-003 produces very few hallucinations under either condition. The authors argue that repeated sampling from one prompt may reproduce a high-probability error, while changing the prompt context can create candidates that avoid it, though the study does not directly demonstrate this proposed change in token distributions.

Overall, the evidence supports three conclusions. MAPS generally improves translation quality under the tested conditions. Its differentiated prompts create a more useful pool than equally sized ordinary reranking. Its final success depends on the selector's ability to manage noisy guidance. The experiments do not establish that the gains will persist with newer models, matched computational budgets, or genuine document-level translation.

4 Related Work and Comparative Context

MAPS belongs to a broader shift from one-step prompting toward structured LLM translation, but related methods step in at different stages: iterative refinement, TasTe, and TEaR all revise an existing translation through drafting, evaluation, and refinement (Chen et al., 2024; Wang et al., 2024; Feng et al., 2025), representing diagnosis and revision more directly than MAPS. MAPS instead prepares before translation, generates parallel alternatives, and selects an existing candidate, and the two approaches are complementary since a MAPS candidate could be selected and then refined.

MAPS's demonstrations are generated for each input and require no retrieval system. This offers inference-time flexibility but provides less grounding than demonstration-aware translation models that retrieve high-quality demonstrations from training data (Li et al., 2024). The same trade-off applies to terminology: self-generated guidance is widely available but less reliable than curated or retrieved resources.

The frameworks also differ in their treatment of multiple hypotheses: MAPS selects one complete candidate, while QE-Fusion combines spans from several translations using quality estimation (Vernikos and Popescu-Belis, 2024), which could preserve complementary strengths (for example, terminology from one candidate and fluency from another) but risks introducing inconsistencies absent from any single output.

Finally, a sentence-level inferred topic is not equivalent to document context. Document-level work supplies surrounding text and evaluates discourse phenomena such as reference, consistency, coherence, and style, and can reduce several error types in paragraph-level literary translation while still producing critical mistakes (Wang et al., 2023; Karpinska and Iyyer, 2023). Empirical translation-process research further shows that human translation involves iterative and concurrent activity rather than a simple fixed sequence (Schaeffer et al., 2016). These comparisons situate MAPS among human-inspired but engineering-oriented translation methods, not models of professional or cognitive translation.

5 Discussion

5.1 Contributions and Methodological Strengths

The strongest contribution of MAPS is structured candidate diversity. Its advantage over an equally sized reranking pool indicates that translation quality depends not only on generating more outputs but also on creating alternatives under meaningfully different conditions, and oracle results suggest that the resulting pool contains stronger candidates than ordinary sampling. This conclusion remains bounded by the experimental control: candidate

quantity and selector are matched, but total computation is not.

A second strength is the framework's treatment of unreliable preparation. MAPS does not combine all guidance into a mandatory prompt or assume that every intermediate output is useful. Separate candidates isolate the effects of different guidance types, and retaining the direct translation protects against harmful generated analyses. The direct candidate's frequent selection and the ablation findings support this design, and MAPS thus manages uncertainty through diversity and evaluation rather than treating model-generated guidance as authoritative.

5.2 Limitations and Unresolved Questions

Candidate selection is both essential and imperfect. COMET-QE is stronger than LLM-SCQ in the reported experiments, but the oracle gap and documented selector errors show that high-quality candidates are sometimes discarded. MAPS consequently needs selectors that are better calibrated for adequacy errors, hallucination, and the particular forms of diversity created by guided prompting.

The second limitation is that all guidance originates from the same underlying model. Keyword pairs, topics, and demonstrations are not independent evidence and may reproduce a shared misconception: the keyword analysis already showed that the model strongly obeys target-side suggestions of uncertain suitability, which cannot be treated like verified terminology or retrieved bilingual evidence. MAPS reduces the consequences of unreliable guidance through selection, but it does not solve this verification problem.

Practical and generalisation evidence is also limited. MAPS requires additional implementation complexity compared with direct translation, and, as noted above, its comparison with reranking is not matched in computation. The experiments are mainly sentence-level and cannot support conclusions about document consistency, discourse, or professional translation factors. Human evaluation covers only English↔Chinese, and the tested systems belong to an earlier generation of LLMs. Evaluation with newer systems, broader professional domains, document-level data, and matched budgets is required before strong deployment claims can be made.

5.3 Extent of Human-Likeness and Future Directions

MAPS is human-like at a limited procedural level: it introduces preparation, alternative generation, and evaluation before final commitment, activities similar to recognisable elements of translation practice. However, the framework does not model attention, cognitive effort, or consultation of trusted resources and experts. Translation-process evidence also suggests that human behaviour is less linear and more iterative than the three-stage architecture implies (Schaeffer et al., 2016).

Improved output quality cannot by itself establish cognitive equivalence. Wu et al. (2025) find no clear evidence that gains from explicit human-like decomposition necessarily arise from that decomposition and show that simpler instructions to translate again can perform competitively or better in their tested settings. This does not undermine MAPS's own controlled reranking evidence for differentiated prompting, but it does weaken any inference from better translations to genuinely human reasoning. MAPS is therefore best understood as a human-inspired engineering framework rather than an empirically validated cognitive model.

Three extensions follow directly from its limitations: activating MAPS adaptively only for sentences with high uncertainty, ambiguity, or terminology difficulty, checking or replacing generated guidance with retrieved terminology, demonstrations, and document context, and making selection an intermediate stage followed by targeted refinement or candidate fusion. These extensions should be evaluated with newer models, stronger human assessment, and comparisons that match candidates, tokens, latency, or financial cost.

6 Conclusion

MAPS shows that LLM translation can improve when inference is organised around structured preparation, differentiated candidates, and explicit evaluation, though its effectiveness remains constrained by imperfect selection, unverified guidance, and limited practical and generalisation evidence.

MAPS is human-like at a limited procedural level, but its broader significance lies in demonstrating that translation systems can benefit from separating interpretation, alternative generation, and final decision-making instead of treating translation as one immediate act of generation.

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